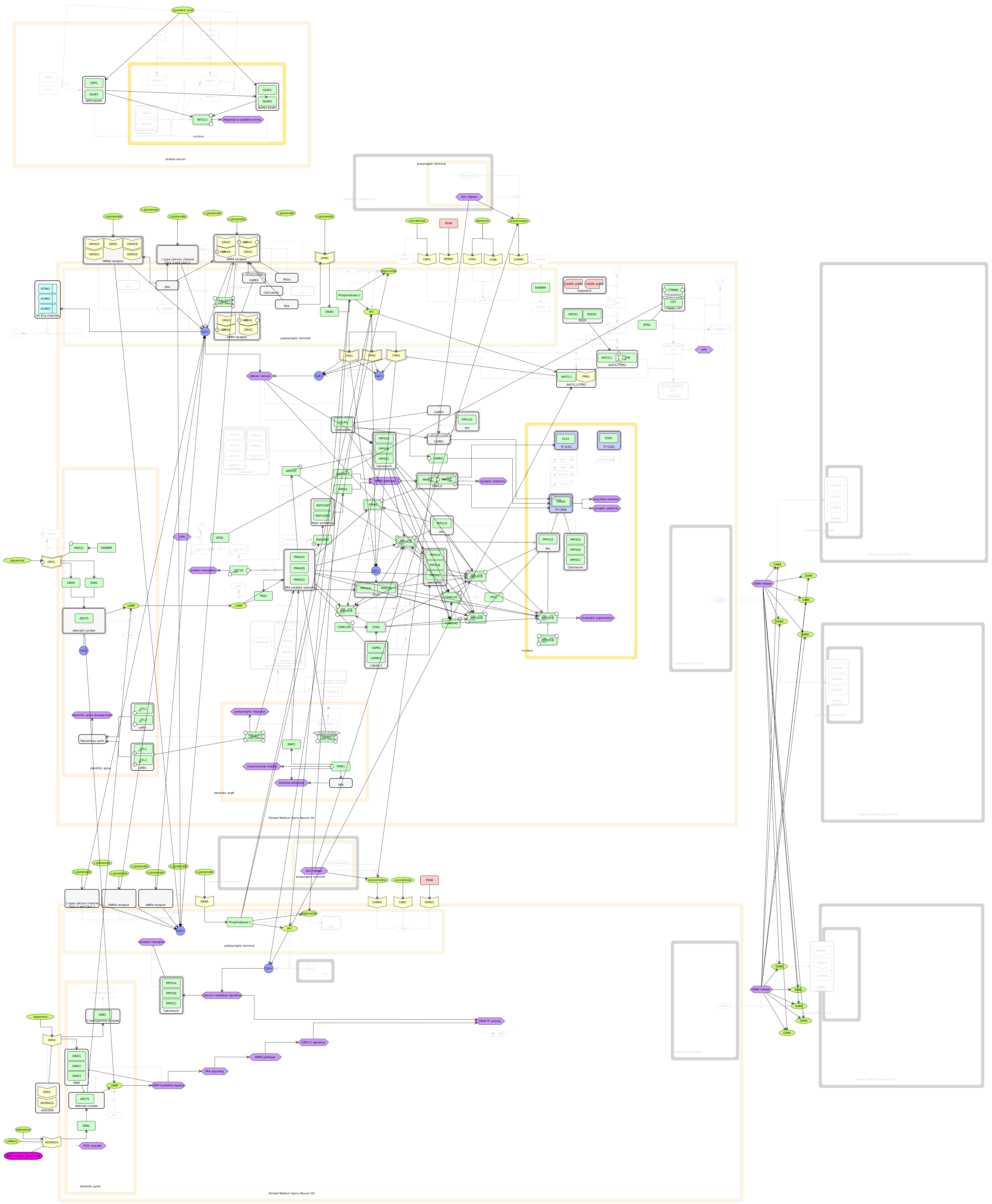
quinolinic acid
NFE2L2
KEAP1
KEAP1
DPP3
DPP3
KEAP1
DPP3:KEAP1
NFE2L2
KEAP1
NUP62
KEAP1
NFE2L2
KEAP1:NFE2L2
NFE2L2
response to oxidative stress
KEAP1
NUP62
NUP62:KEAP1
nucleus
striatal neuron
presynaptic terminal
acetylcholine
cholinergic interneuron
ACh release
L-glutamate
L-glutamate
L-glutamate
L-glutamate
L-glutamate
L-glutamate
L-glutamate
cannabinoid
PENK
serotonin
acetylcholine
GRIN2A
GRIN1
GRIN2B
GRIN2C
GRIN2D
NMDA receptor
L-type calcium channel Cav1.3 and Cav1.2
Ca2+
GRIA3
GRIA1
GRIA4
GRIA2
AMPA receptor
GRM1
TRPV1
CNR1
OPRM1
HTR4
HTR6
CHRM4
NTRK2
KCNN1
KCNN2
KCNN3
SK KCa channels
K+
K+
PP1
PKA
RASGRF1
phosphorylated
CaMKII
Calcineurin
PP2A
PP1
PKA
S935 S1292 S910 S1444 LRRK2
GRIA3
GRIA1
GRIA4
GRIA2
AMPA receptor
Ca2+
postsynaptic terminal
PIP2
diglyceride
Phospholipase C
GNAQ
IP3
cAMP
RANBP9
ITGB1
CASP8 (p18)
CASP8 (p10)
Caspase-8
VIM
ROCK1
ROCK2
ROCK
BTRC
CTNNB1
mutated polyQ
HTT
CTNNB1:HTT
CTNNB1
HTT
UPS
AHCYL1
AHCYL1
S71 S38 VIM
AHCYL:ITPR3
AHCYL1
ITPR3
AHCYL1:ITPR3
HTT
inclusion
ITPR1
ITPR2
ITPR3
Ca2+
Ca2+
cellular calcium
CaMKII
PPP1CA
PP1
CALM1
Calmodulin
phosphorylated
CaMKII
PPP3CA
PPP3CB
PPP3CC
Calcineurin
PPP3CA
PPP3R1
PPP3CB
PPP3R2
PPP3CC
catalytic subunit
regulatory subunit
Calcineurin
CAMK4
ARPP21
S55
RASGRF1
MAPK pathway
RAP1A
MAPK3
MAPK1
T202 Y204
T185 Y187
ERK1/2
synaptic plasticity
ELK1
TF ELK1
EGR1
TF EGR1
EGR1
FOS
DUSP1
PDYN
PPP1R1B
S133 CREB1
TF CREB
long-term memory
synaptic plasticity
PTPN5
RAP1GAP
RAP1GAP2
Rap1 activating
PPP1CA
PP1
PPP1CA
PP1
PPP3CA
PPP3CB
PPP3CC
Calcineurin
DRD1
PRKCD
RANBP9
UPS
BTRC
EEF2K
EEF2K
RASGRP2
PPP1R1B
T34 T75 S137
dopamine
DRD1
GNAS
GNAL
EEF2K
S499
protein translation
PKA catalytic subunit
PRKACA
PRKACB
PRKACG
Ca2+
PPP3CA
PPP3CB
PPP3CC
Calcineurin
PPP1R1B
PP2C
AMP
PDEs
cAMP
cAMP
ADCY5
adenylyl cyclase
ATP
PPP2CA
PPP2CB
PP2A
CSNK1A1
PPP1R1B
CSNK1A1
CDK5
CSNK1A1
PPP1R1B
chromatin organization
PPP1R1B
Ca2+
PRKAR2B
PRKACA
PRKACB
PRKACG
PKA catalytic subunit
LRRK2:PKAR2B:PKAC
CAPN1
CAPNS1
calpain-1
PKA catalytic subunit
PRKAR2B
nucleus
CFL1
CFL2
cofilin
dendritic spine development
filamentous actin
CFL1
CFL2
cofilin
dendritic spine
postsynaptic response
PRKAR2B
S935 S1292 S910 S1444 LRRK2
mutated R1441C
S935 S1292 S910 S1444 LRRK2
MAP2
mitochondrial motility
PINK1
dendrite extension
PKA
dendritic shaft
Striatal Medium Spiny Neuron D1
GABA-A receptor
GABRA1
GABRA2
GABRA3
postsynaptic terminal
substantia nigra pars reticulata
GABA
presynaptic terminal
GABA
GABA release
GABA
GABA
GABA
GABA
GABRA1
GABRA2
GABRA3
GABA-A receptor
postsynaptic terminal
globus pallidus pars interna
cholinergic interneuron
acetylcholine
ACh release
presynaptic terminal
L-glutamate
L-glutamate
L-glutamate
L-glutamate
L-glutamate
L-glutamate
L-glutamate
acetylcholine
cannabinoid
PENK
L-type calcium channel Cav1.3 and Cav1.2
NMDA receptor
AMPA receptor
GRM5
CHRM1
CNR1
OPRD1
Ca2+
PIP2
Phospholipase C
diglyceride
PKC
IP3
cAMP
receptor transport
postsynaptic terminal
Ca2+
Ca2+
ER
Phospholipase C
PPP3CA
PPP3CB
PPP3CC
Calcineurin
Calcium mediated signaling
CREB TF activity
PENK
GNB1
G beta-gamma complex
dopamine
DRD2
ERK1/2 signaling
MAPK pathway
GNAI1
GNAI2
GNAI3
GNAI
PKA signaling
DRD2
ADORA2A
A2A:D2R
cAMP
cAMP-mediated sigaling
ADCY5
adenylyl cyclase
ATP
GNAL
adenosine
caffeine
ADORA2A
PI3K cascade
A2A receptor antagonist
dendritic spine
Striatal Medium Spiny Neuron D2
GABA
GABA release
GABA
GABA
GABA
GABA
presynaptic terminal
GABRA1
GABRA2
GABRA3
GABA-A receptor
postsynaptic terminal
globus pallidus pars externa

ACh release
quinolinic acid
GABA release
DPP3
KEAP1
DPP3:KEAP1
GABA
acetylcholine
PENK
cannabinoid
CHRM1
DRD2
ADORA2A
A2A:D2R
OPRD1
CNR1
adenosine
caffeine
A2A receptor antagonist
dopamine
KEAP1
NUP62
NUP62:KEAP1
serotonin
ROCK1
ROCK2
ROCK
P8 (p18)
CASP8 (p10)
Caspase-8
HTR6
HTR4
BTRC
CHRM4
OPRM1
phosphorylated
CaMKII
RANBP9
Calcineurin
PP2A
S935 S1292
S910 S1444
LRRK2
PKA
GNAQ
diglyceride
KCNN1
KCNN2
KCNN3
SK KCa channels
CNR1
UPS
L-glutamate
ADORA2A
DRD2
NFE2L2
AHCYL1
S71
S38
VIM
AHCYL:ITPR3
CTNNB1
mutated polyQ
HTT
CTNNB1:HTT
GRIA3
GRIA1
GRIA4
GRIA2
AMPA receptor
GRIA3
GRIA4
GRIA
AMPA receptor
GRIN2A
GRIN1
GRIN2B
GRIN2C
GRIN2D
NMDA receptor
GRM1
L-type calcium channel Cav1.3 and Cav1.2
NMDA receptor
AMPA receptor
L-type calcium channel Cav1.3 and Cav1.2
GRM5
GNAL
GNB1
G beta-gamma complex
PI3K cascade
GNAI1
GNAI2
GNAI3
GNAI
response to oxidative stress
PINK1
mutated R1441C
S935 S1292
S910 S1444
LRRK2
AHCYL1
ITPR3
AHCYL1:ITPR3
Ca2+
Phospholipase C
receptor transport
cAMP-mediated sigaling
PKA
MAP2
mitochondrial motility
postsynaptic response
CAPN1
CAPNS1
calpain-1
ITPR1
ITPR2
ITPR3
S3
CFL1
S3
CFL2
cofilin
S3
CFL1
S3
CFL2
cofilin
RANBP9
Ca2+
GNAS
IP3
PKA signaling
dendrite extension
PRKACA
PRKACB
PRKACG
PKA catalytic subunit
CSNK1A1
Ca2+
filamentous actin
PRKCD
ADCY5
adenylyl cyclase
ADCY5
adenylyl cyclase
Ca2+
MAPK pathway
EEF2K
S499
RASGRP2
PDEs
PPP2CA
PPP2CB
PP2A
CDK5
PP2C
S55
ARPP21
dendritic spine development
cAMP
cellular calcium
DRD1
Calcium mediated signaling
ERK1/2 signaling
Striatal Medium Spiny Neuron D2
protein translation
RAP1GAP
RAP1GAP2
Rap1 activating
cAMP
T34 T75
S137
PPP1R1B
CALM1
Calmodulin
PPP3CA
PPP3CB
PPP3CC
Calcineurin
CREB TF activity
PPP3CA
PPP3CB
PPP3CC
Calcineurin
RAP1A
RASGRF1
CaMKII
PPP1CA
PP1
MAPK pathway
T34 T75
S137
PPP1R1B
CSNK1A1
phosphorylated
CaMKII
S133
CREB1
CSNK1A1
T34 T75
S137
PPP1R1B
PPP1CA
PP1
PTPN5
T202
Y204
MAPK3
T185
Y187
MAPK1
ERK1/2
synaptic plasticity
EGR1
TF EGR1
ELK1
TF ELK1
T34 T75
S137
PPP1R1B
T34 T75
S137
PPP1R1B
chromatin organization
PPP3CA
PPP3CB
PPP3CC
Calcineurin
PPP1CA
PP1
CAMK4
TF CREB
synaptic plasticity
long-term memory